

Causal Analysis of Debt-to-Income Ratio on Loan Default Risk: An Instrumental Variables Approach

Wenxuan Zhu, Rajvi Jasani, Yiqiao Zhu
BIOS 681 Group Project Proposal — October 2025

1 Research Question and Background

Our research question asks: *What is the causal effect of a borrower’s debt-to-income ratio (DTI) on the probability of loan default?* Because DTI is likely endogenous—borrowers with higher risk profiles may also have higher DTIs due to unobserved characteristics such as financial behavior or preferential lending terms—ordinary regression estimates would be biased. To address this endogeneity, we employ an instrumental-variables (IV) approach using two potential instruments: (1) `loan_limit` (conforming vs. non-conforming status), which affects maximum borrowing capacity and permissible DTI thresholds, and (2) `approv_in_adv` (pre-approval status), which reflects institutional screening before loan approval. Each instrument provides a different source of quasi-random variation in DTI, allowing us to identify the Local Average Treatment Effect (LATE) for distinct borrower subpopulations and test validity through overidentification tests.

Previous research in credit and econometrics has emphasized the importance of addressing endogeneity when studying loan default behavior. Classic studies on instrumental variables and LATE interpretation (Angrist & Imbens, 1994; Angrist & Pischke, 2009) demonstrate that IV estimates capture causal effects for borrowers whose treatment status is influenced by the instrument. In credit risk contexts, similar IV strategies have isolated the causal impact of borrowing constraints, screening policies, and credit access on default outcomes. By comparing LATEs from two different instruments, we can assess whether both identify consistent causal effects or capture different complier groups, providing insight into heterogeneity in borrowers’ responses to debt burden.

2 Data

The data (<https://www.kaggle.com/datasets/yasserh/loan-default-dataset/data>) comes from a comprehensive bank loan dataset from 2019, originally comprising 148,670 loan records and 34 variables capturing applicant-level and loan-level attributes (loan amount, interest rate, term, DTI, credit score, property value, geographic region, and approval process characteristics). After excluding records with missing values on critical features (`DTI`, `Default Status`, `Credit Score`), the cleaned sample contains 124,547 records, of which 16.3% experienced default and 83.7% were repaid successfully.

2.1 Key Variables

Outcome: `Status` is a binary indicator where 1 denotes default and 0 denotes successful repayment.

Treatment: `dtir1` represents the debt-to-income ratio (continuous), measuring the proportion of income allocated to debt service. According to *A Model of Mortgage Default*, proposed by John Y. Campbell and João F. Cocco, higher DTI suggests greater financial leverage and potentially higher default risk.

Instrument 1: `loan_limit` indicates conforming (`cf`) vs. non-conforming (`ncf`) loans. Conforming loans adhere to regulatory criteria (e.g., Fannie Mae/Freddie Mac standards) that constrain permissible DTI levels, but have no measurable direct effect on default rates once DTI is accounted for, according to *The Influence of Fannie and Freddie on Mortgage Loan Terms* (Alex Kaufman, 2012).

Instrument 2: `approv_in_adv` indicates pre-approval status (`pre` vs. `nopre`). Pre-approval reflects institutional screening influencing acceptable DTI thresholds but should not directly cause default conditional on DTI and other borrower characteristics.

Controls: Credit score, property value, geographic region, loan characteristics (amount, rate, term).

3 Analysis Plan

3.1 Two-Stage Least Squares (2SLS) Estimation

We will use 2SLS to identify the causal effect of DTI on default:

First Stage: Regress DTI on each instrument (separately and jointly) plus controls to establish instrument relevance. We will assess instrument strength using F-statistics (target > 10) and partial R^2 .

Second Stage: Regress default on predicted DTI from the first stage plus controls to obtain LATE estimates.

3.2 Model Specifications

We will estimate three primary models:

1. 2SLS using `loan_limit` only
2. 2SLS using `approv_in_adv` only
3. 2SLS using both instruments jointly

Comparing Models 1 and 2 reveals whether treatment effects differ across complier populations (conforming loan standards vs. pre-approval screening). Model 3 provides efficient estimates when both instruments are valid and enables overidentification testing.

3.3 Validity and Robustness Checks

Instrument Validity:

- First-stage F-statistics and partial R^2 for relevance
- Sargan-Hansen test for overidentification (joint model)
- Covariate balance tests across instrument categories

Sensitivity Analyses:

- E-value calculations to quantify robustness to unmeasured confounding
- Comparison with OLS estimates to assess selection bias magnitude
- Subgroup analyses by credit score (prime vs. subprime) to explore treatment effect heterogeneity
- Alternative specifications (different control sets, functional forms)

This triangulation approach provides credible causal estimates while transparently addressing the strong identifying assumptions required for valid IV inference.

4 Expected Contributions

This project will contribute to consumer credit risk literature by: (1) providing causal evidence on the DTI-default relationship while addressing endogeneity concerns, (2) demonstrating the utility of institutional features as natural instruments for DTI, (3) revealing heterogeneity in treatment effects across complier populations, and (4) informing lending policy by quantifying the marginal impact of DTI restrictions on default risk. Our findings will help financial institutions and regulators better understand the causal mechanisms through which debt burden affects repayment behavior.